

Mapping Refractions

Refractions are used for semi transparent objects to simulate how light bends, or refracts, through a surface. There are two main ways to calculate refractions in the default 3ds Max renderer: with the Raytrace map and the Reflect/Refract map.

The main parameter to use is the index of refraction (IOR) value. This will change, depending on the type of surface. In the Raytrace material, this is adjusted in the map, while in the Raytrace and Reflect/Refract maps, it is adjusted in the Extended Parameters rollout.

3.4 Types of Maps

3ds Max has a wide variety of maps. These fall into two main categories: 2D and 3D maps. 2D maps contain bitmaps and other types of maps that create images defined by two dimensions, and are mapped by using U and V coordinates. 3D maps also have depth and are calculated volumetrically. They're mapped by using UVW coordinates. Maps can also fall into two other categories, bitmaps and procedural:

- Bitmap textures are derived from image or movie files. These can have a wide variety of formats. Because bitmaps are external files, you need to manage these separately.
- Procedural textures create images by using custom algorithms. These can create all sorts of natural looks and phenomena. You can map them to any object by using UV coordinates, and you can also tile them. Procedural textures can be useful because they can create natural-looking effects quickly and with little overhead. You can also animate procedural textures by keyframing their attributes.

3ds Max has a wealth of procedural textures, and you can add to the list by using third-party plug-ins. Here are a few of the more common procedural textures:

Cellular: A map that simulates natural phenomenon such as mosaic tiles, bumpy skin, pebbled surfaces, and more.

Checker: A map that creates a two-colour checkerboard pattern.

Gradient: A map that creates a linear or circular three-color gradient.

Gradient Ramp: A map that creates a user-definable gradient with as many colours as desired.

Noise: A map that creates Gaussian or fractal noise. **Smoke** A map that simulates smoke. It is usually used to create opacity maps for clouds and other smoky effects.

Splat: A map that simulates splattered paint. **Swirl** A map that creates a swirl effect.